جامعة الملك خالد
KING KHALID UNIVERSITY

AI-Enhanced Intrusion Detection System with Risk Assessment

Project (2) -- Final Presentation

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Introduction

Cyber threats are rising with the rapid growth of digital systems, and traditional IDS struggle to detect new or unknown

Traditional IDS systems such as Snort depend on set rules. They can detect known attacks, but they create many alerts that are not organized or prioritized.

This project implements an AI-driven IDS that analyzes network traffic, detects anomalies, and assigns risk scores.

Problem Statement



Traditional IDS systems generate many alerts without proper prioritization. As a result, security teams may waste time analyzing normal alerts while critical attacks receive the same priority level, delaying response to dangerous threats. so, a risk-scoring mechanism is needed to prioritize attacks from highest to lowest severity.

Needs and Design Requirements

Needs	Design requirements
Able to monitor network traffic	• Has Snort IDS installed on Raspberry Pi • Captures real-time network packets
Able to detect and classify threats	• Has Random Forest Machine Learning model • Classifies alerts into Normal, Low, Medium, High, Critical
Able to prioritize and assign risk	• Calculates risk score for each alert • Sorts alerts based on severity to focus on critical threats
Able to alert and notify users	• Has dashboard or interface to display alerts • Can send notifications for high-risk threats
Able to store and process data	• Has storage for logs and network data • Has processing unit to run ML model and Snort efficiently

Design Alternatives

1. **Design Alternative 1:** Traditional Snort Signature-Based IDS
2. **Design Alternative 2:** Snort + Deep Learning on (Jetson Orin)
3. **Design Alternative 3:** Snort + Random Forest + Risk Scoring

Design Alternatives 1

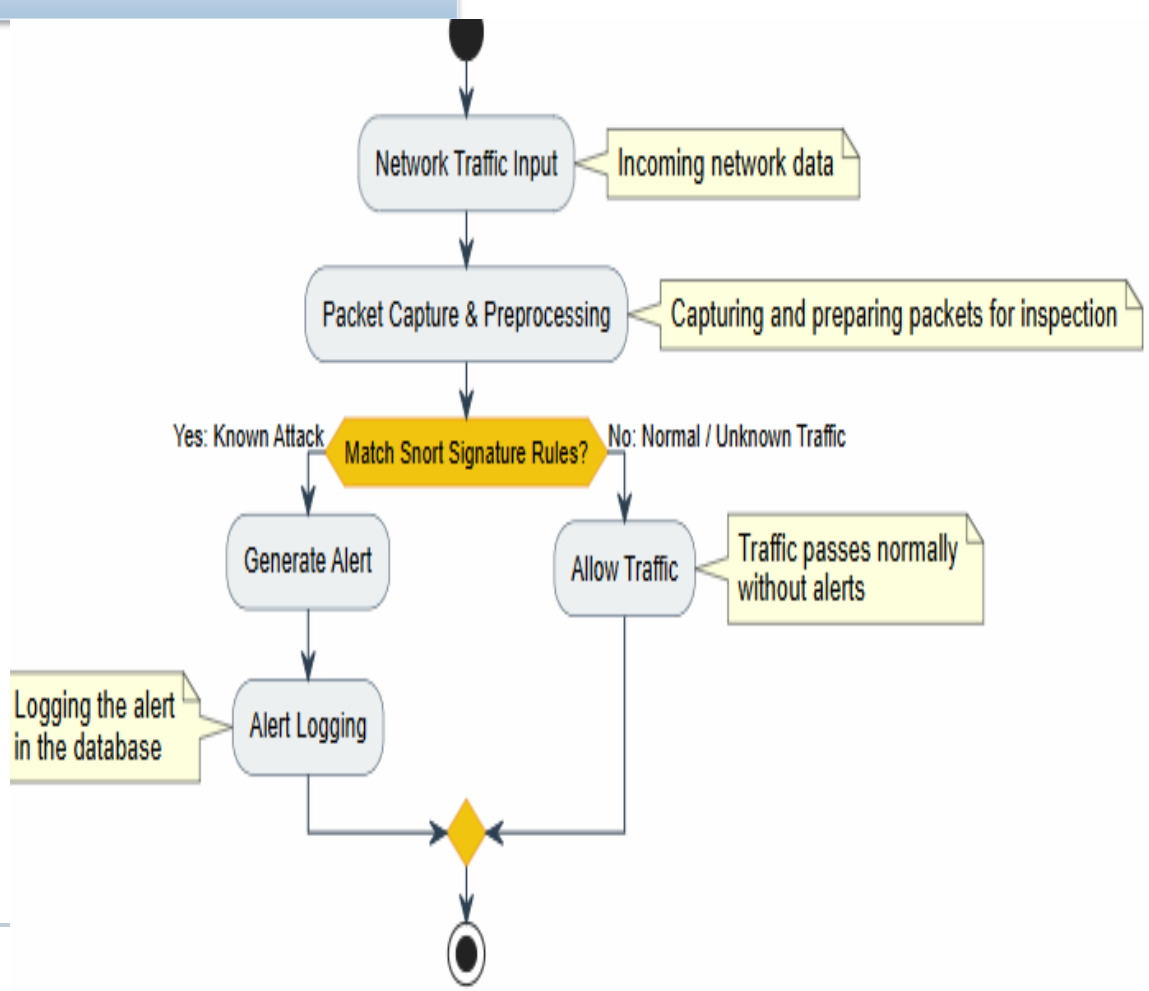
Design Alternatives 1: Traditional Snort Signature-Based IDS

Description:

- Uses Snort only for signature-based detection
- Simple and low-cost
- Good for known attacks

Limitations:

- Cannot detect new or unknown attacks
- Higher false-positive rate
- No risk scoring



Design Alternatives 2

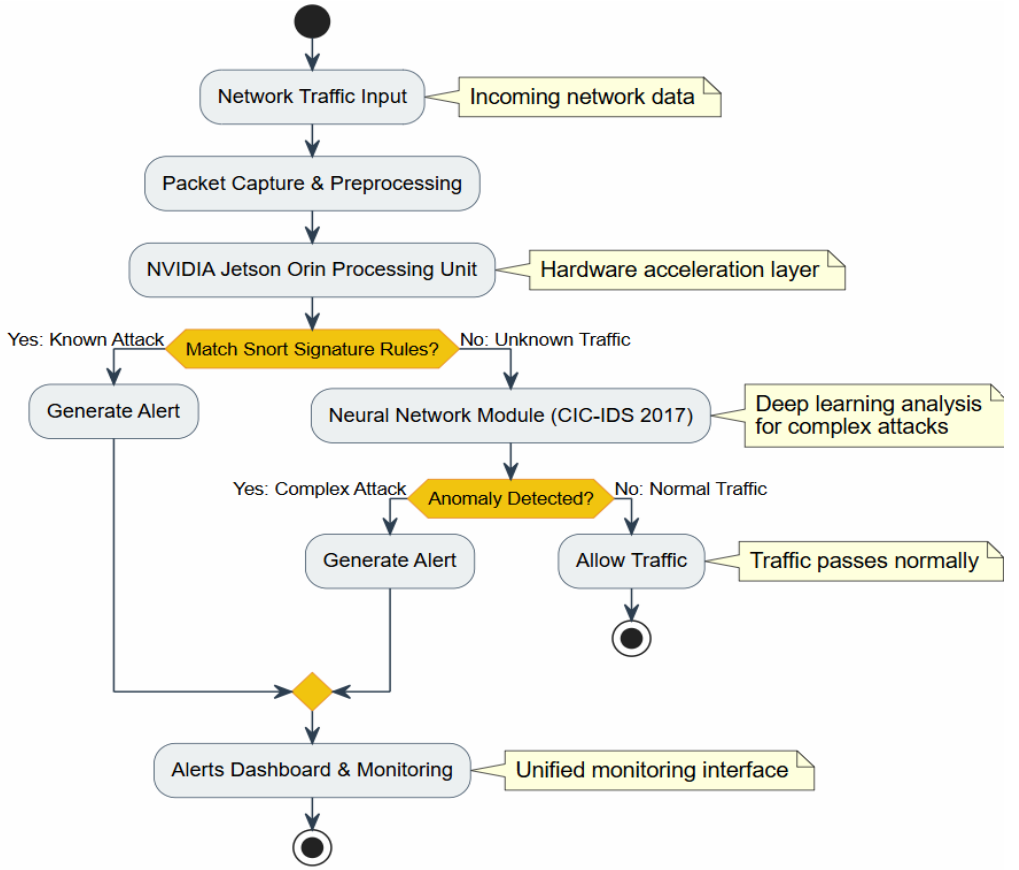
Design Alternatives 2: Snort + Deep Learning on (Jetson Orin)

Description:

- Adds a neural network model trained on CICIDS2017
- Higher detection accuracy
- Better for complex attacks

Limitations:

- Expensive hardware (Jetson Orin)
- Higher power consumption
- Not suitable for small devices



Design Alternatives 3

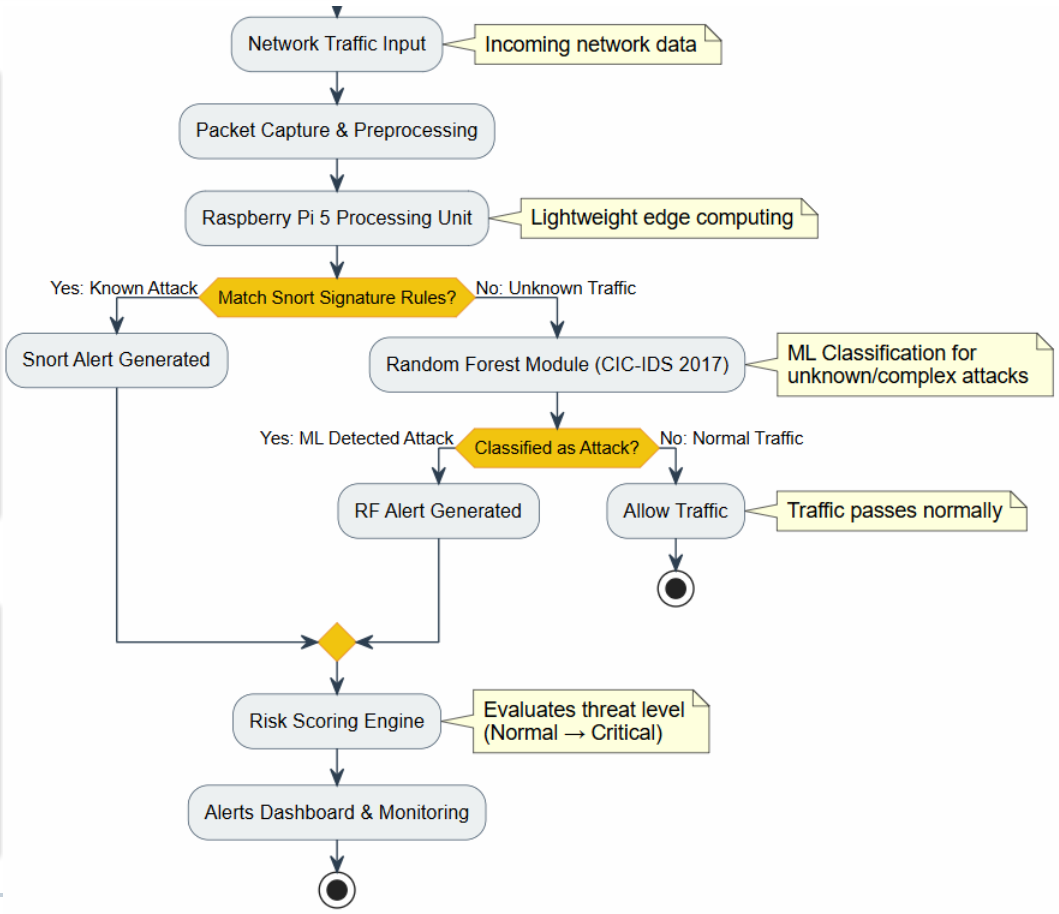
Design Alternatives 3: Snort + Random Forest + Risk Scoring

Description:

- Combines Snort with a Random Forest ML model
- Lower false positives
- Adds risk scoring (Normal → Critical)
- Works well on Raspberry Pi

Limitations:

- Needs model updates
- More complex than Snort-only



Design Alternatives Evaluation Criteria

Design Requirements / Criteria			Design Alternative #1		Design Alternative #2		Design Alternative #3	
	Name	w	v	$w \times v$	v	$w \times v$	v	$w \times v$
Design requirements	Ability to detect attacks	0.20	1	0.20	2	0.40	2	0.40
	Ability to classify threats	0.15	0	0.00	2	0.30	2	0.30
	Ability to prioritize alerts	0.15	0	0.00	0	0.00	2	0.30
	Hardware compatibility	0.10	2	0.20	0	0.00	2	0.20
Criteria	Detection accuracy	0.15	1	0.15	2	0.30	2	0.30
	Cost efficiency	0.10	2	0.20	0	0.00	1	0.10
	Computational efficiency	0.10	2	0.20	0	0.00	2	0.20
	Real-time capability	0.05	2	0.10	1	0.05	2	0.10
Total Points		1	10	1.05	7	1.05	15	1.9

Selected Design Alternative

Selected Design Alternative — Alternative 3

1. Highest evaluation score
2. Detects known and unknown attacks
3. Integrated risk scoring module
4. Real-time alerting dashboard
5. risk scoring help to prioritize threats
6. Supports modern cybersecurity needs
7. Lower false-positive rate compared to Snort-only

Selected Design Alternative

1. IDS Detection:
-Snort Intrusion Detection System (IDS)
-Performs rule-based detection of network attacks

2. Raspberry Pi (Snort IDS):
-Runs the Snort IDS on a Raspberry Pi device
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3. Alert Logs:
-When Snort detects suspicious activity , it generates and sends an alert log

4. Random Forest Model:
-The machine learning model responsible for risk classification
predicts the threat into five risk levels:

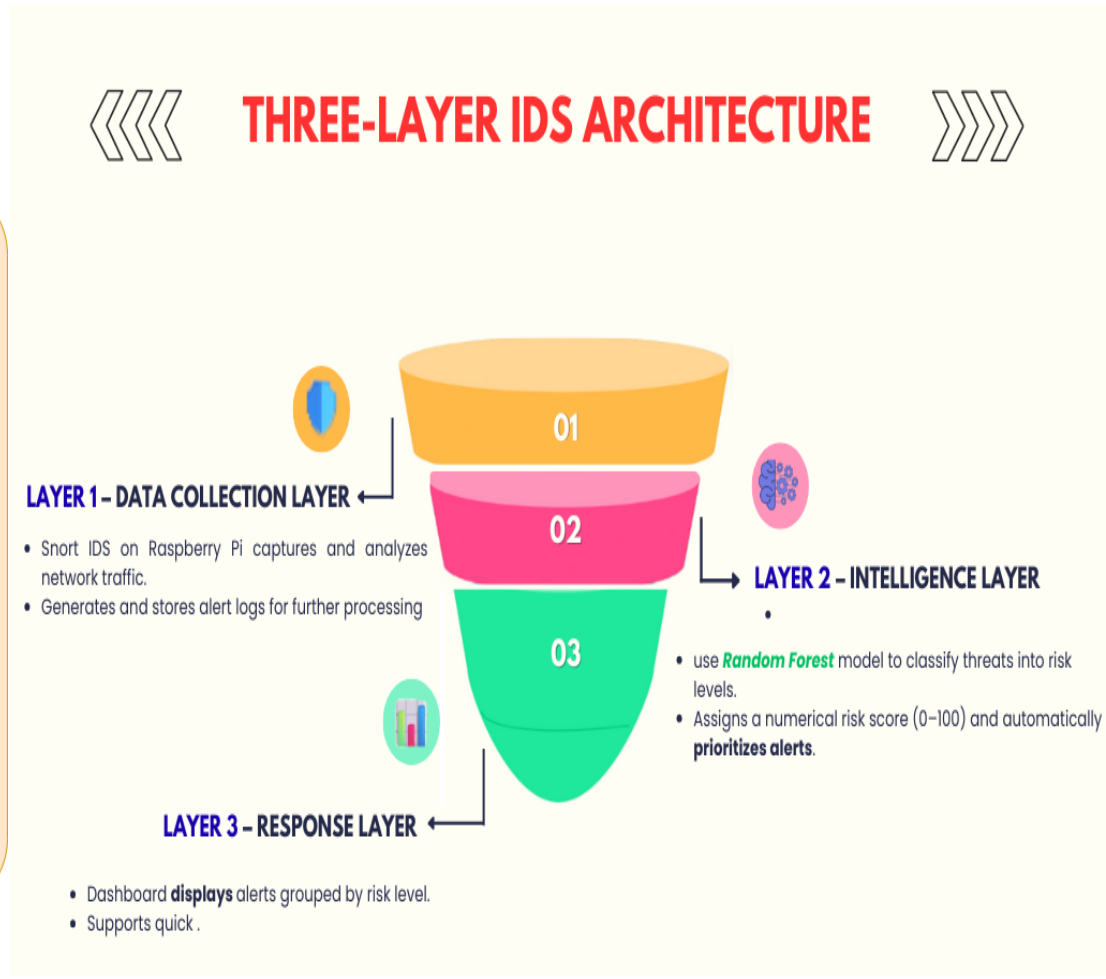
- **Data Collection & Labeling**
  Collect Snort alerts and label them (Normal, Low, Medium, High, Critical).
- **Data Cleaning & Preprocessing**
  Remove duplicates, normalize values, and handle missing data.
- **Feature Engineering**
  Extract features from logs: protocol, port, packet count, bytes, Snort priority, etc.
- **Random Forest Training**
  At this step, the computer learns from the collected data so it can make predictions or decisions automatically in the future.
- **Model Export (Deployed on Raspberry Pi)**
  Save model and preprocessing steps for use in real-time detection.

5. Risk Score:
Assigns a numerical *score* from 0–100 Example: 96 = Critical

6. Risk Assessment Engine (Classified Alert):
-Prioritizes alerts automatically based on the risk score
- Stores classified alerts and risk levels in a central database for analysis

7. Dashboard:
-Displays alert lists grouped by risk level

-Provides recommendations and summaries for security teams



Hardware/Software Requirements

Hardware Component	Description
Raspberry Pi 5	Main device running Snort and ML
MicroSD (256GB)	Stores OS, logs, and datasets
Network Interface	Captures traffic in real time
Power Supply (5V/3A)	Stable power for operation
Protective Case	Protects the Raspberry Pi
Cooling Fan(optional)	Keeps temperature safe

Hardware/Software Requirements

Software	Description
Snort IDS	Real-time packet inspection
Python 3	ML and data processing
Scikit-Learn	Random Forest model
Ubuntu OS	Runs all system software
Flask Dashboard	Shows alerts and risk levels

Related Standards

Category	Standard	Application
Ethernet	IEEE 802.3	Wired network compatibility
Wi-Fi	IEEE 802.11	Wireless traffic support
OS	POSIX	Ensures system compatibility

Realistic Constraints and Metrics

Type	Item	Description
Constraint	**Economic**	Low project cost
Constraint	**Environmental**	Low power usage
Constraint	**Ethical**	Respect user privacy
Constraint	**Political**	Comply with local regulations
Constraint	**Physical**	Limited Pi resources
Metric	**Accuracy**	Attack detection level
Metric	**False Positives**	Wrong alerts rate
Metric	**Resource Usage**	CPU/RAM load
Metric	**Latency**	Processing delay
Metric	**Risk Score Quality**	Correct severity level

Detailed Design Description

- Snort is used to analyze network traffic
- Alerts are collected and processed
- A machine learning model is used for classification
- A risk score is assigned to each alert
- A dashboard is used to display the results

Detailed Design Description

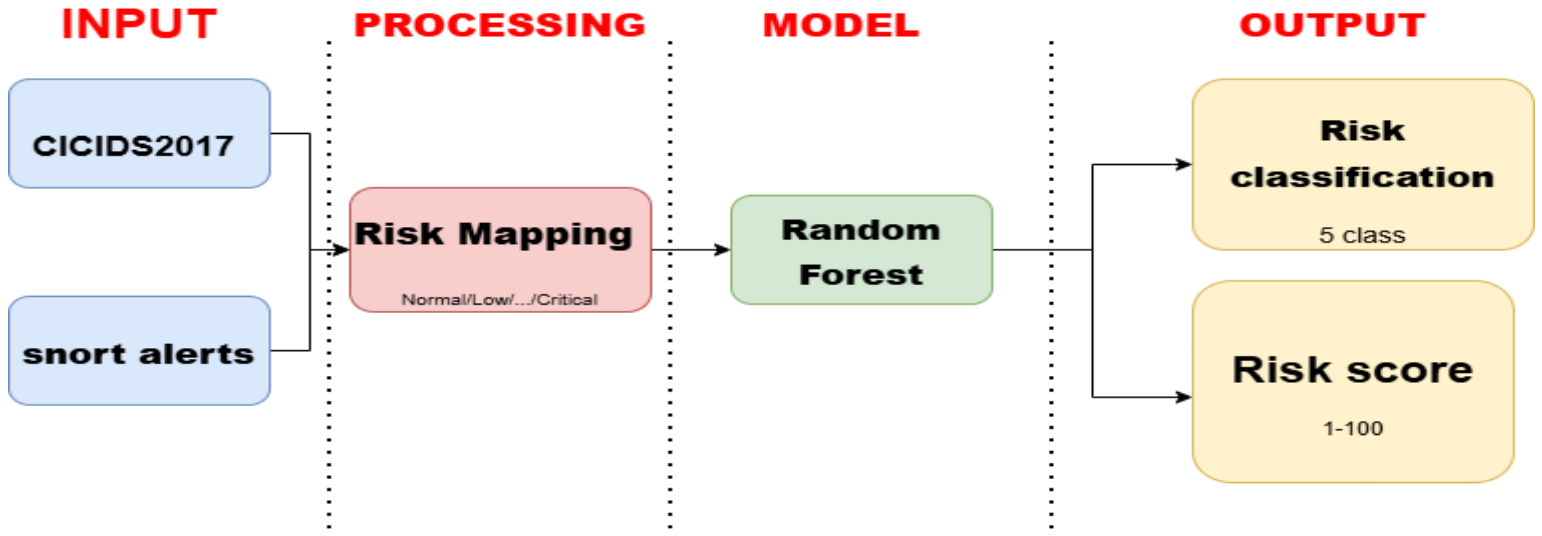
CICIDS2017 dataset was used

Data preprocessing was applied

Random Forest was used as the model

The system classifies normal and attack traffic

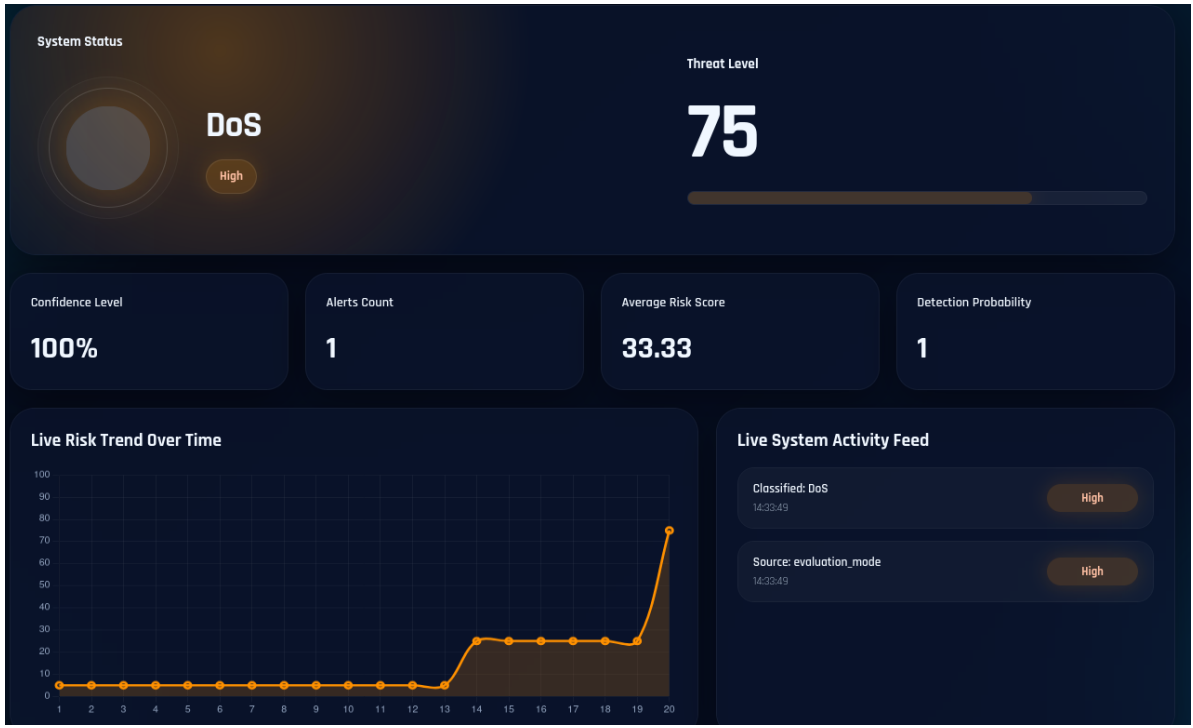
The output is converted into five risk levels



Modeling and/or Simulation (If applicable)

- **Demo:**





Implementation (Software and/or Hardware)

- Installed Snort IDS on Raspberry Pi 5
- Used Python and Scikit-Learn for ML integration
- Trained Random Forest model using CICIDS2017 dataset
- Processed Snort alerts and assigned risk scores
- Displayed alerts using Flask dashboard

System Testing, Results and Discussion

	precision	recall	f1-score	support
0	1.00	1.00	1.00	454620
1	1.00	1.00	1.00	2767
2	1.00	1.00	1.00	76138
3	0.99	1.00	0.99	31786
4	0.93	0.77	0.84	402
5	1.00	0.96	0.98	436
accuracy			1.00	566149
macro avg	0.99	0.95	0.97	566149
weighted avg	1.00	1.00	1.00	566149

- The system was tested using simulated network traffic and attack scenarios
- Snort generated alerts which were analyzed using the Random Forest model
- The model achieved an overall accuracy close to 99%
- Most attack classes achieved high precision and recall values
- The results demonstrate effective attack classification performance
- The system successfully classified attacks into different risk levels in real time

Conclusions and Recommendations

Successfully integrated Snort with Random Forest ML

Improved intrusion detection and alert prioritization

Low-cost and lightweight IDS solution

Suitable for small networks and educational environments

Future work: add deep learning and automatic response features

References

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THE END

Thank You